# Science

Chemistry Lessons
Chemistry Labs
Nature Notebook
Nature Walks & Scouting

Alveary

# Science: Chemistry

GRADE(S): 9-12+

## About the Course

This course includes the following topic(s): Chemistry Lessons, Chemistry Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Chemistry Lessons
Alveary High School Chemistry provides a natural next step following Physical Science (e.g., Alveary Form 3 Science) and establishes a solid foundation for High School Biology. The course progresses through all expected topics in introductory chemistry and incorporates living engagement and special attention to citizenship, problem-solving, and communication skills for a complete Charlotte Mason science course. This course includes our video companion series, currently at no additional cost.

About Chemistry Labs
Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.

About Nature Notebook: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

## Placement & Combining Tips

Chemistry Lessons
Recommended for learners who have completed Physical Science, such as Alveary Grade 7-8 Science, including the laboratory activities. Learners should be taking at least Algebra 1 alongside Chemistry (Algebra 2 if using Denison Success). Teachers wishing to place students in Physical Science instead of Chemistry may choose either to complete Alveary Grade 7-8 Science in a single year (with 5 lessons + 1 lab each week available in the Grade 7-8 Quick Links) or to purchase separately from Classical Academic Press the Novare Introductory Physics Program and Video Course.

Nature Notebook: Grades 9-12
Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12
Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

## Scheduling

| GRADE | SCHEDULE INFO. | BOOKS |
|---|---|---|
| 9-12+ | Chemistry Lessons<br>5 times/week 45 min | HS Chem Intro<br>Introductory Chemistry: A Foundation<br>Michael Faraday's The Chemical History of a Candle<br>The Chemical Elements Coloring and Activity Book |
| 9-12+ | Chemistry Labs<br>1 time/week 60 min | Alveary Chemistry Lab Book |
| 9-12 | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+ | |
| 9-12 | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ | |

Sample Weekly View

| Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|---|---|---|---|---|
| **Science: Chemistry** | | | | |
| Chemistry Lessons | Chemistry Lessons<br>Nature Walks & Scouting: Grades 1-8 | Chemistry Lessons | Chemistry Lessons<br>Nature Notebook: Grades 9-12 | Chemistry Lessons<br>Chemistry Labs |

## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Chemistry Lessons
☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

☐ Obtain materials from the supply lists.

☐ Plan for discussion and engagement. Teachers who do not feel that they can discuss the subject of this course with students should plan to watch lesson videos, preread the material, or complete the course alongside their students. Expertise is not necessary, but discussion is as important in math and science as in history and literature.

☐ Labs/activities are presented in a fully integrated state in which they are connected to the lessons. To use them in this way, it is helpful for the lab to be scheduled after the 5th lesson. If your lab time is earlier in the week and you will find it difficult to be flexible, plan to delay the start of the lesson schedule so that lab day occurs after the 5th lesson. For example, if your lab time is on Tuesday, you might begin lesson 1 on Wednesday of the first week of school rather than on Monday. Those wanting to use the labs/activities on a more flexible schedule can do so, but should be ready and able to adjust lesson connections.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

### Term Prep & Teacher Tips

Chemistry Lessons
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- 3 small objects from around the house
- distilled or deionized water
- mineral spirits/paint thinner (may substitute mineral oil or cooking oil, but won't be able to evaporate the solvent at the end of the lab)
- 4 recycled soda cans
- sticky tack, a piece of foam, or anything to hold the thermometer in a soda can and close the opening
- hydrogen peroxide antiseptic
- barometer or computer access to check the local weather station
- ice
- sugar
- salt
- spoon
- lemon juice
- stopwatch, such as those found on most electronic devices
- microwave-safe bowl
- microwave
- whisk that you do not care about
- another whisk or electric mixer
- well-ventilated space (porch, carport, etc)
- juice or tea
- quart-sized mason jar with a lid
- jar, cup, or bowl
- slotted spoon, strainer, or coffee filter
- 2 large egg whites (Term 3)
- large mixing bowl
- cookie sheet
- oven
- optional: hard blunt tip (knitting needle, skewer, etc)
- optional: 227g coconut oil from the grocery store, IF you want to use the soap you will make
- optional: lemon-lime soda, ice cream, yogurt, or cottage cheese (Term 3)
- optional: cream of tartar
- optional: piping bag
- optional: food coloring
- optional: parchment paper

### Reminders

Chemistry Lessons

☐ Note that egg whites are used in the last lab of the year. Make a note in your calendar if you will need to purchase these closer to the appropriate time. Refer to the lab book for more details.

## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

**Science: Chemistry**

### Chemistry Lessons

HS Chem Intro

Introductory Chemistry: A Foundation

Michael Faraday's The Chemical History of a Candle

The Chemical Elements Coloring and Activity Book

### Chemistry Labs

Alveary Chemistry Lab Book

## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- View Basic Supplies
- View Supply List Details

**Science: Chemistry**

Alveary High School Chemistry Kit

Food-Grade Calcium Chloride

Food-Grade Sodium Alginate

Food-Grade Sodium Hydroxide

Household Items - Science: Chemistry

Organic Chemistry Model Kit

Scale/Balance

### Chemistry Labs

Safety Goggles

## Quick Links

Click THIS text or scan the QR code for links.

**Science: Chemistry**

- Chemistry Lab Book

### Chemistry Lessons

- Alveary Bookshelf
- Contact Us for Teachers
- Homework Help for HS Science Students
- John Muir Laws • Nature Journaling Resources
- Khan Academy Video Playlists
- Lab Notebook Examples

- ∞ Seek app from iNaturalist
- ∞ SkyView Lite for Android
- ∞ SkyView Lite for iOS

  

# Science: Chemistry

## How To Approach

### Introduce

• Learners begin each lesson with their existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to consider what learners think about it, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Take note of the chapter or section heading, and consider how the day's topic connects back to prior topic(s).

• Look back as needed. The outline of the book and even of the chapter is helpful in drawing out and connecting ideas. Learners practice making these connections as they proceed through the book.

### Read

• Read or do as instructed in the lessons. Teachers should note any Teacher Tips provided.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.

• If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab book further supports major concepts from the text.

### Narrate

• Process the ideas of the lesson by explaining a concept, describing an object, retelling events, etc. Consider what the point of the lesson is before deciding what to do with it.

• Learners may use words, pictures, models, graphics, etc, to process and convey ideas in their own way.

### Discuss

• Consider together any thoughts, confusion, or concerns about the passage.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there are any dates to keep for the Book of Centuries or quotes for the Commonplace Book.

## Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc, depending on learner needs and interest.

# Science: Chemistry

## How To Approach

### Introduce

- Before beginning, be sure that the basic rules of lab safety (as listed in the lab manual) are understood and obeyed.

- Every science lab has the same flow, which follows the scientific method and is guided by the lab book.

- Learners begin with an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.

- If learners are unsure what to do with the Introduction, it can help to dialogue with someone as they work through the ideas. Check the Chemistry Companion in the Quick Links if a video tutorial might be helpful. This year is a pilot for this video series, so videos will be added to the Quick Link throughout the year.

- Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas and hypotheses into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.

- If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or use of assistive technology, as appropriate.

- After they complete their prelab narration, learners should collect the listed materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.

### Lab Procedure

- Learners complete the procedure as instructed in the lab book.

- The lab gives instructions for using the lab notebook to create tables and diagrams.

### Analysis & Conclusions

- The last step in the lab is to analyze the data and observations and draw conclusions from them. This postlab narration is prompted in the lab manual. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about, if they were to continue.

- Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

- Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we

| | |
|---|---|
| | could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned. |

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 1**

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

PREP: Be sure you have read the Course Notes and Planning and Prep. Read Teacher Tip

**→ INTRODUCTION**

Watch the course introduction video:

∞ Video Link: HS Chemistry Intro (10:02)

In the 1840s, Michael Faraday began a public lecture series for young people, called the Chemical History of the Candle. The lecture series is an excellent demonstration of the scientific method and many fundamental concepts in chemistry. As you watch the lectures this week, create an outline and record any points that stand out to you. This will help you to identify the main points and connect them. You can use these notes to support narration.

**→ ACTIVITY**

Cut out the spiral from Lab 1 and attach a piece of thread by pushing a threaded needle through the dot in the center; observe the candle, recording observations in your notebook. Light the candle and observe. Using the thread, hold the spiral over the candle (without allowing it to catch fire) and observe. Blow out the candle and observe.

∞ Image Link: Lab Notebook Samples

∞ Video Link: Paper Spiral Demo (1:56)

∞ Image Link: Most Lab 1 materials

**→ VIEW, NARRATE, & DISCUSS**

Lecture 1

∞ Video Link: Engineering Guy #2 (11:47)

**→ ACTIVITY**

Complete the follow-up activity in your lab book.

**● PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

★ TEACHER TIP

The linked videos are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! We especially recommend watching the course introduction video with students!

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WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 2**

Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

Prep: Read Teacher Tip

**→ NOTE**

Use the prompts bulleted in the lessons to help you engage with the material and to self-assess and decide if you are ready to move on. You don't need to be an expert in any or all of the topics, but the ideas should feel familiar, and you should be able to think about and discuss the ideas.

★ TEACHER TIP

The discussion prompts bulleted in the lessons provide teachers with helpful ways to engage even when students are working independently.

## Term 1

**→ VIEW, NARRATE, & DISCUSS**
Lecture 2
∞ Video Link: Engineering Guy #3 (13:44)

• Discuss how Faraday's demo indicates the presence of tiny particles in the candle's flame and how these tiny particles produce the brilliance of the flame.
• What can be concluded about the nature of the particles, and what is still in question regarding combustion production? Consider: Does combustion produce a pure substance or a mixture of substances that can be detected and separated? How did Faraday conclude that the black particles are C? How and what did Faraday observe about the invisible particles?

**→ ACTIVITY**
Complete the follow-up activity in your lab book.

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WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 3**

▯ Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

**→ VIEW, NARRATE, & DISCUSS**
Lecture 3
∞ Video Link: Engineering Guy #4 (15:42)

• Compare/contrast the three phases observed as the candle burns with the three phases of water.
• Explain why water is still a pure substance even though it can be divided into two parts.
• How do oxygen and hydrogen relate to the combustion reaction?

★ STUDENT TIP
Come to science ready to work a puzzle. If you're missing a piece of that puzzle, don't wait—message Homework Help for Students!

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WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 4**

▯ Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

**→ VIEW, NARRATE, & DISCUSS**
Lecture 4
∞ Video Link: Engineering Guy #5 (20:00)

• Discuss why it is important that we understand that air is a mixture rather than a pure substance or nothing at all.
• Add to your previous diagram or make a new one, indicating what we now know about the products of the combustion reaction: C soot from the candle, water made of H from the candle, and O from the air, and something else.
• Describe the characteristics of the unknown substance.

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

**→ ACTIVITY**
Complete the follow-up activity in your lab book.

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 5**

▯ Materials: The Chemical History of the Candle Video link

High School Chemistry Lab Book and materials listed within

**→ VIEW, NARRATE, & DISCUSS**
Lecture 5
∞ Video Link: Engineering Guy #6 (21:06)

- How does Faraday demonstrate that the invisible substance is carbon dioxide? How does he show that carbon dioxide is a compounded substance? What is it made of?
- Discuss the uniqueness of carbon-based fuels.
- Describe respiration/breathing in terms of combustion and explain the resulting interdependence of plants and animals.

WEEK 1 ⬚ **60m Chemistry Lessons - Lesson 6**

*Observing a Candle*

▯ Materials: High School Chemistry Lab Book and materials listed within

**→ LAB DAY**
Complete the Analysis and Conclusions for Lab 1. Explore Chapter 3 of the Introductory Chemistry text if you need help.
∞ Video Link: Lab 1 Conclusion (6:37)

WEEK 2 ⬚ **45m Chemistry Lessons - Lesson 7**

▯ Materials: Introductory Chemistry

Prep: Read Teacher Tip

**→ INTRODUCTION**
Just as you did with the Faraday lectures, continue to outline or diagram your readings, noticing main points and recording any definitions or important ideas. Even though you are externalizing the ideas while you do the lesson, you should still narrate and discuss at the end of the lesson. Review your notes at least weekly throughout the year.

Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

★ TEACHER TIP
The linked videos are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there is an alternative in the Quick Links.

  

Links, as well as the Homework Help link to message Mrs. Merritt.

**→ READ, NARRATE, & DISCUSS**

Ch.1

∞ Video Link: Scientific Method (6:08)

- What is Chemistry?
- Describe 'the Scientific Method.'
- Discuss the Critical Thinking question in Sec.1-4.
- Discuss Active Learning #6.

WEEK 2 ⬚ **45m Chemistry Lessons - Lesson 8**

Materials: Introductory Chemistry

**→ NOTE**

When you complete exercises for the chapter, try a selection from different sections to help discern where your understanding is secure and where you may need some additional review and practice. The problems suggested below are simply those that seemed interesting to us; you may choose different ones. Those demonstrated in the video are noted. The point is not necessarily to get them all correct right away; sometimes exercises can stimulate new thoughts on the ideas. The point is to find where you are comfortable and where you may be confused. If the exercises seem comfortable, then continue. If they seem confusing or too challenging, then review the associated reading(s) and practice some more. This self-assessment is essential as you continue your education. The answers to even-numbered exercises are provided in the back of the book, so that you can check your understanding. Note that books always have some typos; however, don't be too dependent on the answers in the back!

**→ REVIEW AND PRACTICE**

Ch.1 Review and Suggested Exercises:
Act. #12, 13,
1.2 #5, 6,
1.3 #7,
1.4 #10, 13,
1.5 #17

∞ Video Link: Chapter 1 Practice (8:35).
Act. #12, 13,
1.3 #7, edition 8e

WEEK 2 ⬚ **45m Chemistry Lessons - Lesson 9**

Materials: Introductory Chemistry

**→ NOTE**

Always be sure to follow the example problems in the text and complete the self-check exercises as you read. If that means

! STUDENT REMINDER
**Remember:** scientific thinking is like solving a puzzle. If you're missing a piece of that puzzle, don't wait–message Homework Help for Students!

## Term 1

proceeding at a slower pace and spreading lessons over more than one day, that is fine.

If you want to go over zero and negative exponents, consider watching the Math Break video:
∞ Video Link: Math Break: Zero and Negative Exponents (3:48)

**→ READ, NARRATE, & DISCUSS**
Ch.2.1-2.4
∞ Video Link: Scientific Notation (8:45)

- Why do we use scientific notation, and how does it work?
- What is 'uncertainty' in a measurement, and how do significant figures help?
- Discuss the Critical Thinking question in Sec.2-2.

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WEEK 2

**45m Chemistry Lessons - Lesson 10**

Materials: Introductory Chemistry

**→ NOTE**
There is a misprint in this section for the 9th edition. The fractions printed in blue are missing their bar/line.

**→ READ, NARRATE, & DISCUSS**
Ch.2.5-2.6
∞ Video Link: Significant Figures (10:46)
∞ Video Link: Unit Conversions (4:59)

- Discuss Active Learning #7.
- What is a conversion factor? What is its value? Why must this always be true?
- Explain how units help you check your math.

! STUDENT REMINDER
Don't forget to review your lesson notes!

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WEEK 2

**45m Chemistry Lessons - Lesson 11**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.2.7-2.8
∞ Video Link: Temperature and Density Problems (8:33)

- Explain the differences between the three temperature scales.
- What does it mean to say that something is dense?

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[Click THIS text or scan the QR code for links.]

## Term 1

WEEK 2

⬚ **60m Chemistry Lessons - Lesson 12**

*Evaluating Density*

▯ Materials: High School Chemistry Lab Book and materials listed within

Prep: Read Teacher Tip

**→ LAB DAY**

Complete Lab 2.

∞ Video Link: Density Lab (3:55)

∞ Image Link: Examples of Lab 2 materials

★ TEACHER TIP

Asking students to explain what they did in the lab is a great way to begin a discussion! Follow it up with, "Why/how did you do that?"

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 13**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Ch.2 Review and Suggested Exercises:
Act.#1-5,
2.1 #7, 8,
2.2 #16,
2.3 #22, 24,
2.4 #30, 32,
2.5 #33, 48, 50,
2.6 #64, 66,
2.7 #74,
2.8 #90, 94

∞ Video Link: Chapter 2 Practice (33:07).
Act. #1-4,
2.1 #7, 8,
2.3 #22, 24,
2.5 #33, 48, 50,
2.6 #64,
2.7 #74,
2.8 #90, 94 from 8e

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**

⬚ nature walk - record observations
⬚ science free read

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 14**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Continue exercises.

● Remember to self-assess! If the exercises made sense, then move on. If they seemed confusing, then go back to the associated readings or use Homework Help.

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 15**

▯ Materials: Introductory Chemistry

High School Chemistry Lab Book

PREP: Read Teacher Tip

**➔ REVIEW**

Briefly review the definition you wrote in the Analysis and Conclusions of Lab 1.

**➔ READ, NARRATE, & DISCUSS**

Ch.4.1-4.4: As you visualize the atom, model it with the parts of your model kit, by sketching it, or with any medium you like, such as modeling clay or Lego!

∞ Video Link: Elements (10:05)

- What is an element?
- Tell what you know about Dalton's theory of atoms.
- What do we mean by the 'law of constant composition'?
- Explain what a chemical formula is and how chemical formulae work.
- Get to know your model kit by building the molecules in example 4.1 (or any suggested in your kit). You can use your kit to help you visualize molecules anytime you want throughout the course.

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 16**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.4.5-4.7

∞ Video Link: Atoms and Isotopes (8:43)

- What do you know about the parts of atoms?
- How did scientists learn about the parts of the atom?
- Choose at least one Critical Thinking question in Sec.4-5 to discuss.
- What is an isotope?

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 17**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.4.8-4.9

∞ Video Link: Meeting the Table (5:55)

- Tell about or diagram what you know about the periodic table so far.
- Discuss the differences between different elements, such as gold and hydrogen.
- Are elements atoms or molecules? Explain.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 1

**→ SUPPLEMENTAL**
Check out some other ways to visualize the periodic table:
∞ Image Link: The Periodic Table in 3D
∞ Image Link: Cylinder with Bulges

WEEK 3

⬚ **60m Chemistry Lessons - Lesson 18**

> ▯ Materials: Coloring book or any blank periodic table

PREP: Read Teacher Tip

**→ ACTIVITY**
Use what you have learned to color your own copy of the Periodic Table.

★ TEACHER TIP
Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking? Do they know how to find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-solving? If you are unsure about their progress, reach out through Contact Us.

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 19**

> ▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.4.10-4.11
∞ Video Link: Ions (9:42)

- What is an ion?
- How does the periodic table predict ions?
- Tell what you know about ionic compounds.
- Discuss the Critical Thinking question in Sec.4-11.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 20**

> ▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.4 Review and Suggested Exercises:
Act.#12, 14,
4.1 #4, 5,
4.2 #10-13,
4.4 #20,
4.6 #24, 26, 28,
4.7 #36, 38, 42,
4.8 #44, 48, 49, 53,
4.9 #56, 58, 60, 64,
4.10 #66, 68, 70, 74, 76, 78,
4.11 #80, 82, 84

∞ Video Link: Chapter 4 Practice (13:56).
4.2 #11,
4.4 #20,

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

## Term 1

4.7 #38, 42,
4.10 #74,
4.11 #84, from 8e

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WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 21**

Materials: Introductory Chemistry

**➜ REVIEW AND PRACTICE**
Continue exercises.

• Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER
Don't forget to review your lesson notes!

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WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 22**

Materials: Coloring book

**➜ ACTIVITY**
If you finished your exercises, then explore the coloring book to begin learning about some elements that you are interested in.

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WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 23**

Materials: Coloring book

Prep: Read Teacher Tip

**➜ ACTIVITY**
Continue exploring elements or choose a game from the back to learn.

★ TEACHER TIP
Ask students to tell you about an element or to teach you a game from the back of the coloring book!

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WEEK 4 ⬚ **60m Chemistry Lessons - Lesson 24**

Materials: Coloring book

**➜ ACTIVITY**
Continue exploring elements or learn a game from the coloring book.

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WEEK 5 ⬚ **45m Chemistry Lessons - Lesson 25**

Materials: Introductory Chemistry

**➜ READ, NARRATE, & DISCUSS**
Ch.5.1-5.2
∞ Video Link: Naming Ionic Compounds (7:33)

• Using a periodic table, how would you name CsBr? Which is the positively charged cation and which is the negatively charged anion? For each, what is their charge, and how many are in the compound?
• Explain the difference between iron (II) chloride and iron (III) chloride. Which is the positively charged cation and which is the

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

## Term 1

| | | |
|---|---|---|
| | negatively charged anion? For each, what is their charge, and how many are in the compound?<br>● Discuss the Critical Thinking question in Sec.5-2. | |
| WEEK 5 | ⬚ **45m Chemistry Lessons - Lesson 26**<br>▯ Materials: Introductory Chemistry<br><br>PREP: Read Teacher Tip<br><br>**➔ READ, NARRATE, & DISCUSS**<br>Ch.5.3-5.4<br>∞ Video Link: Naming Nonmetal Compounds (4:31)<br><br>● How would you name a compound with two nitrogen atoms and four oxygen atoms?<br>● How would you write the formula for phosphorus pentachloride?<br>● Use your model kit to build the compounds in example 5.5. | ★ TEACHER TIP<br>The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters! |
| WEEK 5 | ⬚ **45m Chemistry Lessons - Lesson 27**<br>▯ Materials: Introductory Chemistry<br><br>**➔ READ, NARRATE, & DISCUSS**<br>Ch.5.5-5.7<br>∞ Video Link: Naming Polyatomic Ions and Acids (8:32)<br><br>● What do we mean by "polyatomic ions"?<br>● In Na2SO4, which part is the cation and which part is the anion?<br>● Tell what you know about acids and how we name them.<br>● Discuss the Critical Thinking question in Sec.5-7.<br>● Use your model kit to build the polyatomic ions in Table 5.4. | ! STUDENT REMINDER<br>Don't forget to review your lesson notes! |
| WEEK 5 | ⬚ **45m Chemistry Lessons - Lesson 28**<br>▯ Materials: Introductory Chemistry<br><br>**➔ REVIEW AND PRACTICE**<br>Ch.5 Review and Suggested Exercises:<br>Act.#4, 6,<br>5.2 #10, 14, 16,<br>5.3 #18,<br>5.5 #30, 34, 36,<br>5.6 #40,<br>5.7 #42, 44, 46, 50, 52<br><br>∞ Video Link: Chapter 5 Practice (7:20).<br>Act. #6,<br>5.7 #50, 52, from 8e | ! STUDENT REMINDER<br>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset! |

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

WEEK 5

⬚ **45m Chemistry Lessons - Lesson 29**

*Separating Iodine from a Mixture*

Materials: High School Chemistry Lab Book and materials listed within

**➔ READ & NARRATE**

Read the Introduction to Lab 3 and complete the introductory narration in your notebook.

∞ Video Link: Separation Lab (9:45)

∞ Image Link: Most Lab 3 materials

WEEK 5

⬚ **60m Chemistry Lessons - Lesson 30**

*Separating Iodine from a Mixture*

Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**➔ LAB DAY**

Complete Lab 3.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 31**

Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**

Continue exercises.

● Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 32**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ INTRODUCTION**

Recall what you have learned so far about the signs of a chemical reaction.

**➔ READ, NARRATE, & DISCUSS**

Ch.6.1-6.2

∞ Video Link: Chemical Reactions (5:35)

● What kinds of observations indicate a chemical reaction?

● Explain the meaning of the following chemical equation: $H_2 + O_2 \rightarrow H_2O$

● Use your model kit to build the reactants for the reaction above. Try

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

to rearrange the atoms to make the product. What do you notice? Try the reaction in section 6.2, if you like.
● What does it mean for a chemical equation to be balanced? Why must it be balanced?

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 33**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.6.3: If helpful, use your model kit to build the equation as you read to visualize how it's balanced. Use your kit like this any time you want.
∞ Video Link: Balancing Equations (5:33)

● Explain how to balance a chemical equation.
● Choose at least one Critical Thinking question in Sec.6-3 to discuss.

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 34**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.6 Review and Suggested Exercises:
Act.#4, 7, 14,
6.1 #1, 6,
6.2 #14, 18, 24, 34,
6.3 #35, 40, 43

∞ Video Link: Chapter 6 Practice (46:34).
Act. #7, 14,
6.1 #1,
6.2 #14, 24, 34,
6.3 #40, 43, from edition 9e

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 35**

▯ Materials: High School Chemistry Lab Book

Link

**→ REPORT WRITING**
Begin drafting a lab report for the iodine extraction lab. This is simply a formal report based on what is already in your lab notebook. Similar to the linked sample, include a heading with the experiment title, course name, and your name and date; an Introduction (a more polished version of your introductory narration); Methods (this is your procedure or any important information that someone repeating your experiment would need to know); Results (this is your data, including any tables, graphs, or photographs that your want to provide); a Discussion or Data Analysis; and a Conclusion. The last two sections are a more polished version of your analysis and concluding narration.
∞ Link: Sample Lab Report

! STUDENT REMINDER
Don't forget to review your lesson notes!

# Science: Chemistry

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## Term 1

WEEK 6 ⬚ **60m Chemistry Lessons - Lesson 36**

Materials: High School Chemistry Lab Book

**➔ REPORT WRITING**
Complete your lab report. Review and edit it to make improvements. While editing, use complete sentences and avoid using the first person. In technical writing, such as this, the sentence "I tied the string to the end of the meter stick." should be rephrased, "The string was tied to the end of the meter stick."
∞ Image Link: Use Passive Voice

WEEK 7 ⬚ **45m Chemistry Lessons - Lesson 37**

Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**
Continue exercises.

● Think you're ready to move on? Remember to take a moment for self-assessment!

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 7 ⬚ **45m Chemistry Lessons - Lesson 38**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ READ, NARRATE, & DISCUSS**
Ch.7.1-7.2
∞ Video Link: Precipitation Reactions (6:08)

● Describe some factors that cause a reaction to tend to occur. Do you think these are connected to the signs of a reaction in the last chapter?
● Describe what happens in a precipitation reaction, such as when silver nitrate and potassium chloride are mixed in aqueous solution. Use a sketch or diagram, if helpful.
● Why is the formation of precipitate evidence of a chemical reaction? Use a sketch or diagram, if helpful.
● Discuss the Critical Thinking question in Sec.7-2.

★ TEACHER TIP
The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 7 ⬚ **45m Chemistry Lessons - Lesson 39**

Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.7.3-7.4
∞ Video Link: Acid-Base Reactions (Reactions Producing Liquid Water) (6:13)

! STUDENT REMINDER
Don't forget to review your lesson notes!

# Science: Chemistry

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## Term 1

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| | • What are acids and bases?<br>• What do we mean by "strong" acids and bases?<br>• Explain what is produced when strong acids and bases react? | |
| WEEK 7 | ⬚ **45m Chemistry Lessons - Lesson 40**<br><br>▯ Materials: Introductory Chemistry<br><br>**➔ READ, NARRATE, & DISCUSS**<br>Ch.7.5<br>∞ Video Link: Redox Reactions (Reactions Transferring Electrons) (4:00)<br><br>• What do we mean by "oxidation-reduction"? Use a sketch or diagram, if helpful.<br>• If an element is one of the products or reactants, then the reaction is oxidation-reduction. Discuss this statement. | ! STUDENT REMINDER<br>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset! |
| WEEK 7 | ⬚ **45m Chemistry Lessons - Lesson 41**<br>*Observing Microscale Reactions*<br><br>▯ Materials: High School Chemistry Lab Book and materials listed within<br><br>**➔ READ & NARRATE**<br>Read the Introduction to Lab 4 and complete the introductory narration in your notebook.<br>∞ Video Link: Microscale Reactions Lab (5:22)<br>∞ Image Link: Most Lab 4 materials | |
| WEEK 7 | ⬚ **60m Chemistry Lessons - Lesson 42**<br>*Observing Microscale Reactions*<br><br>▯ Materials: High School Chemistry Lab Book and materials listed within<br><br>PREP: Read Teacher Tip<br><br>**➔ LAB DAY**<br>Complete Lab 4. | ★ TEACHER TIP<br>Ask students to explain their lab to you! |
| WEEK 8 | ⬚ **45m Chemistry Lessons - Lesson 43**<br><br>▯ Materials: Introductory Chemistry<br><br>**➔ NOTE**<br>These sections are about classifying the reactions we have been talking about. The new terms are something to think about, but do not spend time memorizing different types of reactions. They are more for your awareness.<br><br>**➔ READ, NARRATE, & DISCUSS** | ! STUDENT REMINDER<br>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.<br><br>• **PLAN WEEKLY**<br>⬚ nature walk - record observations<br>⬚ science free read |

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## Term 1

Ch.7.6-7.7

● Look back at the objectives for these sections. Discuss the ideas.
● Discuss the Critical Thinking question in Sec.7-7.

WEEK 8 ⬚ **45m Chemistry Lessons - Lesson 44**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.7 Review and Suggested Exercises:
Act.#2, 4, 14,
7.1 #2,
7.2 #6, 10, 18, 22,
7.3 #24, 28,
7.4 #32, 36, 40,
7.5 #44, 46, 50,
7.6 #52, 54,
7.7 #60, 62

∞ Video Link: Chapter 7 Practice (29:38).
Act. #2,
7.2 #18,
7.3 #28,
7.4 #40,
7.6 #54, from 9e

WEEK 8 ⬚ **45m Chemistry Lessons - Lesson 45**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Continue exercises.

● How is your self-assessment going? If you need help, click on Homework Help!

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 8 ⬚ **45m Chemistry Lessons - Lesson 46**

Materials: Introductory Chemistry

Prep: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.8.1-8.2
∞ Video Link: Atomic Mass Units (4:41)

**→ NOTICE THAT THE DISCUSSION PROMPTS COME FROM THE OBJECTIVES AT THE START OF ASSIGNED SECTIONS. THESE OBJECTIVES HELP YOU KNOW WHAT YOU SHOULD BE ABLE TO NARRATE AND DISCUSS. YOU DO NOT NEED TO HAVE FORMULAE AND PROCEDURES MEMORIZED, BUT RATHER BE**

★ TEACHER TIP
Note that the discussion prompts bulleted in the lessons come from the objectives at the start of the assigned sections. These objectives provide teachers with helpful ways to engage, even when students are working independently.

# Science: Chemistry

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## Term 1

**ABLE TO DISCUSS THEM. IF THE IDEAS SEEM FAMILIAR ENOUGH TO NARRATE AND DISCUSS, THEN CONTINUE. IF THEY SEEM CONFUSING OR TOO CHALLENGING, THEN REVIEW THE ASSOCIATED READING(S) OR SEND A MESSAGE TO HOMEWORK HELP. THESE SELF-ASSESSMENT SKILLS WILL HELP YOU AS YOU PROGRESS IN YOUR EDUCATION.**

- What do we mean by "average mass"? Explain how this idea can be used to count something.
- Explain atomic mass. How is it determined?
- What are "atomic mass units"?

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WEEK 8

⬚ **45m Chemistry Lessons - Lesson 47**

> Materials: Introductory Chemistry

Website link

**➜ READ, NARRATE, & DISCUSS**

Ch.8.3-8.4

∞ Video Link: The Mole (4:22)

- Use the idea of Avogadro's number to explain what a mole is.
- Explain how one converts between moles, mass, and number of atoms. Diagram or demonstrate, as needed.
- Tell about the process of problem-solving. How do you do it?
- Choose at least one Critical Thinking question in Sec.8-3 to discuss.
- Did you know that some unit definitions have recently changed? The values have not changed much, but the way they are standardized has been updated. Check out the reasons and the changes at The New SI link.

∞ Website Link: The New SI

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

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WEEK 8

⬚ **60m Chemistry Lessons - Lesson 48**

*Counting by Weighing*

> Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**➜ LAB DAY**

Complete Lab 5.

∞ Video Link: Counting by Weighing Lab (2:04)

∞ Image Link: Most Lab 5 materials

★ TEACHER TIP

Ask students to explain their lab to you!

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WEEK 9

⬚ **45m Chemistry Lessons - Lesson 49**

> Materials: Introductory Chemistry

**➜ READ, NARRATE, & DISCUSS**

Ch.8.5-8.6

∞ Video Link: Molar Mass and Percent Composition (9:58)

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**• PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 1

- What is molar mass?
- Explain how one converts between moles and mass of a sample of a chemical compound. Diagram or demonstrate, as needed.
- What do we mean by "mass percent"?

WEEK 9

**45m Chemistry Lessons - Lesson 50**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.8.7-8.9: If helpful, use your model kit to collect the correct ratio of atoms indicated, to visualize multiples of this ratio, and to experiment with how the atoms might fit together.

∞ Video Link: Empirical and Molecular Formulae (8:15)

- What do we mean by "empirical formula"?
- Discuss how empirical formulae are determined.
- How can an empirical formula help us get to a molecular formula?
- Discuss the Critical Thinking question in Sec.8-8.

★ TEACHER TIP

The videos and discussion prompts are a great way to keep up and engage with students even when they are working independently. Your positive support matters!

WEEK 9

**45m Chemistry Lessons - Lesson 51**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Ch.8 Review and Suggested Exercises:

Act. #4, 17,
8.2 #4, 6, 8,
8.3 #10, 12, 18, 24,
8.5 #26, 30, 34, 38,
8.6 #48,
8.7 #54,
8.8 #60, 66,
8.9 #76, 78

∞ Video Link: Chapter 8 Practice (40:35).

Act. #4, 17,
8.2 #8,
8.3 #18,
8.5 #34, from 9e

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 9

**45m Chemistry Lessons - Lesson 52**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Continue exercises.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 9 ⬚ **45m Chemistry Lessons - Lesson 53**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.9.1-9.2: If helpful, use the model kit to build the water molecules on the reactant side of the equation. Use only those atoms to build the products. Use the model kit for any of the reactions you want to visualize in this way.

∞ Video Link: Stoichiometry Part 1 (3:03)

● What kind of information does a chemical equation provide?

● How do we relate moles of reactants and products using a chemical equation?

WEEK 9 ⬚ **60m Chemistry Lessons - Lesson 54**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.9.3: If helpful, use the model kit to represent the propane and oxygen reactants in example 9.4. Use only those atoms and all of those atoms to try to build the products. Do you have too much or too little of something? How might you resolve the discrepancy? Use the model kit for any of the reactions you want to visualize in this way.

∞ Video Link: Stoichiometry Part 2 (5:42)

● How do we relate masses of reactants and products in a chemical equation?

● Discuss the Critical Thinking question in Sec.9-3.

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 55**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.9.4-9.5

∞ Video Link: Limiting Reagents (6:54)

● What do we mean by "limiting reagent" or "limiting reactant"?

● Explain how one can identify a limiting reagent?

● How does a limiting reagent affect stoichiometric calculations?

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 56**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.9.6
∞ Video Link: Percent Yield (2:52)

- Discuss actual yield versus theoretical yield.
- Explain how actual yield can be determined from theoretical yield.

**→ REVIEW AND PRACTICE**
Ch.9 Review and Suggested Exercises:
Act. #2, 6,
9.1 #2, 4,
9.2 #12, 16,
9.3 #19, 21, 30, 38,
9.5 #52, 57,
9.6 #62, 65

∞ Video Link: Chapter 9 Practice (31:01).
Act. #2, 6,
9.3 #19, 21,
9.5 #57,
9.6 #65, from 9e

★ TEACHER TIP
The videos and discussion prompts are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 57**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Continue exercises.

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 58**

Materials: Current Events link

Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below:
∞ Website Link: Science News for Students
∞ Website Link: Chemistry at Science Daily
∞ Website Link: Chemical & Engineering News

**→ REVIEW AND PRACTICE**
Continue exercises.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 1

• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 59**

*Evaluating Limiting Reagents*

Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**

Read the Introduction to Lab 6 and complete the introductory narration in your notebook.

∞ Video Link: Limiting Reagents Lab (1:10)

∞ Image Link: Most Lab 6 materials

WEEK 10 ⬚ **60m Chemistry Lessons - Lesson 60**

*Evaluating Limiting Reagents*

Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**

Complete Lab 6.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 61**

Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**

Spend this week writing a lab report for Lab 6, then review what you have learned so far this term.

• Review and edit the lab report before turning it in. Use complete sentences, good grammar, third-person voice, and significant figures and scientific notation, as appropriate.

• As you look back over your notes and narrate objectives from the chapters, reflect on whether ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**• PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 62**

Materials: High School Chemistry Lab Book

Introductory Chemistry

PREP: Read Teacher Tip

★ TEACHER TIP

When student(s) take exams next week, evaluate what they recall, but more importantly whether they can explain their thinking and demonstrate a habit of problem-solving.  If no relational growth is observed, consider why: do they need

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 1

| | | |
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| | **→ REPORT WRITING & REVIEW**<br>Spend this week writing a lab report for Lab 6, then review what you have learned so far this term. | more practice, more support, less business, etc? Reach out through Contact Us, if you need help. |
| WEEK 11 | **45m Chemistry Lessons - Lesson 63**<br>Materials: High School Chemistry Lab Book<br>Introductory Chemistry<br><br>**→ REPORT WRITING & REVIEW**<br>Spend this week writing a lab report for Lab 6, then review what you have learned so far this term. | |
| WEEK 11 | **45m Chemistry Lessons - Lesson 64**<br>Materials: High School Chemistry Lab Book<br>Introductory Chemistry<br><br>**→ REPORT WRITING & REVIEW**<br>Spend this week writing a lab report for Lab 6, then review what you have learned so far this term. | |
| WEEK 11 | **45m Chemistry Lessons - Lesson 65**<br>Materials: High School Chemistry Lab Book<br>Introductory Chemistry<br><br>**→ REPORT WRITING & REVIEW**<br>Spend this week writing a lab report for Lab 6, then review what you have learned so far this term. | |
| WEEK 11 | **60m Chemistry Lessons - Lesson 66**<br>Materials: High School Chemistry Lab Book<br>Introductory Chemistry<br><br>**→ REPORT WRITING & REVIEW**<br>Spend this week writing a lab report for Lab 6, then review what you have learned so far this term. | |
| WEEK 12 | **45m Chemistry Lessons - Lesson 67**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed. | |

# Science: Chemistry

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## Term 1

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| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 68**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 69**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 70**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 71**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **60m Chemistry Lessons - Lesson 72**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |

# Science: Chemistry

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## Term 2

WEEK 1

⬚ **45m Chemistry Lessons - Lesson 73**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ REMINDER**
Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick Links, as well as the Homework Help link to message Mrs. Merritt.

**➔ READ, NARRATE, & DISCUSS**
Ch.10.1-10.3
∞ Video Link: Energy and Heat (7:39)

- Discuss "energy," types of energy, and conservation of energy.
- Explain the difference between temperature and heat.
- Discuss the direction of energy flow in a system.
- Discuss the Critical Thinking question in Sec.10-1.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

★ TEACHER TIP
The linked videos and the objectives at the start of each section are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there are some alternatives in the Quick Links.

WEEK 1

⬚ **45m Chemistry Lessons - Lesson 74**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.10.4-10.5
∞ Video Link: Thermodynamics (8:24)

- Explain how the internal energy (E) of a system is affected when heat flows into the system. When heat flows out of the system?
- Discuss the measurement of heat.
- Discuss the Critical Thinking question in Sec.10-4.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 1

⬚ **45m Chemistry Lessons - Lesson 75**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.10.6-10.7
∞ Video Link: Enthalpy (4:05)

**➔ RECALL THAT THESE DISCUSSION PROMPTS COME FROM THE OBJECTIVES AT THE START OF ASSIGNED SECTIONS. IN THE NEXT FEW WEEKS, THE PROMPTS RELATED TO THOSE OBJECTIVES WON'T BE IN YOUR LESSON PLANS, SO PRACTICE USING THEM IN THE TEXT FOR YOUR SELF-ASSESSMENT. IF**

# Science: Chemistry

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## Term 2

**THE IDEAS SEEM FAMILIAR ENOUGH TO NARRATE AND DISCUSS, THEN CONTINUE. IF THEY SEEM CONFUSING OR TOO CHALLENGING, THEN REVIEW THE ASSOCIATED READING(S) OR SEND A MESSAGE TO HOMEWORK HELP.**

- Explain what you know about "enthalpy."
- Explain Hess's Law, relating to enthalpy and reactions that occur in steps.
- Discuss the Critical Thinking question in Sec.10-7.

---

WEEK 1

⬚ **45m Chemistry Lessons - Lesson 76**

Materials: Introductory Chemistry

**→ CATCH UP & REVIEW**

Complete any unfinished work, then do a cumulative review of all you have learned, including what you learned in Term 1.

---

WEEK 1

⬚ **45m Chemistry Lessons - Lesson 77**

*Finding Enthalpy*

Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**

Read the Introduction to Lab 7 and complete the introductory narration in your notebook.

∞ Video Link: Enthalpy Lab (3:50)

∞ Image Link: Most Lab 7 materials

---

WEEK 1

⬚ **60m Chemistry Lessons - Lesson 78**

*Finding Enthalpy*

Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**

Complete Lab 7.

★ TEACHER TIP

Ask students to explain their lab to you!

---

WEEK 2

⬚ **45m Chemistry Lessons - Lesson 79**

Materials: Introductory Chemistry

Prep: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.10.8-10.9

∞ Video Link: Energy and Us (4:55)

- What is meant by the "quality" of energy? How does this change as

**• PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

★ TEACHER TIP

Use the objectives at the start of assigned sections to engage in discussion and note progress even when students are working independently. Students should begin using those objectives to assess their

# Science: Chemistry

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## Term 2

| | | |
|---|---|---|
| | energy is used?<br>• Discuss what you know about types of energy resources.<br>• Discuss the Critical Thinking question in Sec.10-9. | learning over the next few weeks, if they aren't already doing so. |
| WEEK 2 | **45m Chemistry Lessons - Lesson 80**<br>Materials: Introductory Chemistry<br><br>**→ READ, NARRATE, & DISCUSS**<br>Ch.10.10<br>∞ Video Link: Entropy (2:43)<br><br>• Discuss energy as a "driving force" for natural processes.<br><br>**→ REVIEW AND PRACTICE**<br>Ch.10 Review and Suggested Exercises:<br>Act. #2, 9,<br>10.2 #8,<br>10.3 #12, 14,<br>10.4 #17, 20,<br>10.5 #22, 24, 30, 32,<br>10.6 #42,<br>10.7 #48,<br>10.9 #58,<br>10.10 #64<br><br>∞ Video Link: Chapter 10 Practice (21:22).<br>Act. #9,<br>10.4 #17, 20,<br>10.5 #24, 32,<br>10.6 #42,<br>10.7 #48, from 9e | ! STUDENT REMINDER<br>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion. |
| WEEK 2 | **45m Chemistry Lessons - Lesson 81**<br>Materials: Introductory Chemistry<br><br>**→ REVIEW AND PRACTICE**<br>Continue exercises.<br><br>• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing? | ! STUDENT REMINDER<br>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset! |
| WEEK 2 | **45m Chemistry Lessons - Lesson 82**<br>Materials: Current Events link<br><br>**→ INTRODUCTION**<br>Not all chemists agree about the role that entropy plays in the universe. Physical Chemist Arieh Ben-Naim describes entropy as "one of the most misunderstood concepts in physics" and argues that scientists "do not have license to predict the death of the universe..." | ! STUDENT REMINDER<br>Don't forget to review your lesson notes! |

## Term 2

You will have the opportunity to learn much more about entropy in future coursework. While scientists continue to research and strive to understand thermodynamics better, we can still seek ways to use energy responsibly and steward well all that we have been given. The linked article, "Energy Current Event," will give you one example of this complex challenge.

**→ READ, DISCUSS, & NARRATE**
∞ Article Link: Energy Current Event

WEEK 2 ⬚ **45m Chemistry Lessons - Lesson 83**

Materials: Internet or library access

PREP: Read Teacher Tip

**→ INTRODUCTION**
As citizens, we need to think about how to solve problems, like how we generate and use energy. These kinds of problems can be difficult because there are rarely good or bad solutions. We want to make decisions that do the most good and the least harm, and sometimes, the best solution may be different in different communities. You will have three periods for this research and will prepare a composition about your findings later.

**→ RESEARCH**
Learn about the life cycle of one traditional energy source in your area, such as fossil fuels, and one alternative energy source, such as solar, wind, nuclear, or fuel cell technology. For each energy source, try to answer these questions:
- What raw materials are needed to support the technology? How and from where are those materials obtained?
- How is the energy produced and distributed?
- What kind of waste is produced by using this energy source?
- How long do the technology and its support structures last? What kind of waste needs to be disposed of?
- How does this energy source positively and negatively affect people and the environment?
- What systems in society support the use of this energy source (e.g., economic, transportation, etc)?

★ TEACHER TIP
Ask students about their citizen science project. What are they researching? How are they planning to present their findings?

WEEK 2 ⬚ **60m Chemistry Lessons - Lesson 84**

Materials: Internet or library access

**→ RESEARCH**
Continue researching energy sources.

# Science: Chemistry

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## Term 2

**WEEK 3**

☐ **45m Chemistry Lessons - Lesson 85**

Materials: Internet or library access

**➜ RESEARCH**
Complete your research. Use any remaining time to begin thinking about your composition. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully! You could compose an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**• PLAN WEEKLY**
☐ nature walk - record observations
☐ science free read

**WEEK 3**

☐ **45m Chemistry Lessons - Lesson 86**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➜ READ, NARRATE, & DISCUSS**
Ch.11.1-11.4: Just like in Ch.4, model what you are learning about the atom, so that you can visualize it.
∞ Video Link: Light Energy (3:49)

• Discuss Rutherford's nuclear model of the atom.
• Tell what you know about electromagnetic radiation.
• Explain how atoms emit light.
• Discuss how emission spectra demonstrate that light is quantized. Use a diagram, if helpful.
• Discuss the Critical Thinking question in Sec.11-4.

★ TEACHER TIP
The videos and discussion prompts/section objectives are a great way to keep up and engage with students even when they are working independently. Your positive support matters!

**WEEK 3**

☐ **45m Chemistry Lessons - Lesson 87**

Materials: Introductory Chemistry

Website link

**➜ READ, NARRATE, & DISCUSS**
Ch.11.5-11.7
∞ Video Link: Updating the Atom (best with full screen) (6:04)

• Discuss Bohr's model with circular orbits. Sketch or diagram, if helpful.
• Explain what you know about the wave mechanical model. Sketch or diagram, if helpful.
• Tell what you know about s, p, and d orbital shapes. Sketch or diagram, if helpful.
• For a 3D visualization of the orbitals described in section 11.7, check out "hydrogenic atomic orbitals" at the link. You can manipulate and rotate them to see them better.
∞ Website Link: Orbital Visualizations

! STUDENT REMINDER
Don't forget to review your lesson notes!

# Science: Chemistry

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## Term 2

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 88**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.11.8-11.9
∞ Video Link: Electron Configuration (best with full screen) (9:00)

● Tell what you know about energy levels and orbitals in the wave mechanical model. Sketch or diagram, if helpful.
● Tell what you know about electron spin. Sketch or diagram, if helpful.
● Tell what you know about how electrons fill energy levels in atoms larger than hydrogen. Sketch or diagram, if helpful.
● Discuss valence versus core electrons.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 89**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.11.10
∞ Video Link: Electron Configuration Continued (6:38)

● Explain what you know about electron configuration in atoms with atomic numbers greater than 18.
● Discuss the Critical Thinking question in Sec.11-10.

WEEK 3

⬚ **60m Chemistry Lessons - Lesson 90**

*Observing the Chemistry of Electronic Configuration*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete Lab 8.
∞ Video Link: Electron Configuration Lab (2:01)
∞ Image Link: Most Lab 8 materials

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 91**

▯ Materials: Introductory Chemistry

Coloring book

**→ READ, NARRATE, & DISCUSS**
Ch.11.11
∞ Video Link: Periodic Trends (4:06)

● Using a periodic table, describe trends in atomic properties.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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## Term 2

→ **ACTIVITY**
Color additional copies of the Periodic Table to illustrate the periodic trends you have learned about.

WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 92**

▯ Materials: Introductory Chemistry

Prep: Read Teacher Tip

→ **NOTE**
Now that you've got the hang of using the chapter exercises to help self-assess your learning, you should choose some exercises for yourself. Don't forget to consider whether they feel comfortable or confusing before deciding whether to move on or practice more. Remember that the point is not necessarily to get them all correct right away; sometimes exercises can stimulate new thoughts on the ideas. The point is to find where you are comfortable and where you may be confused.

→ **REVIEW AND PRACTICE**
Ch.11 Review and Suggested Exercises:
11.1 #2,
choose 2 from 11.2,
choose 1 from 11.3,
choose 2 from 11.4,
11.6 #26,
choose 3 from 11.7,
11.9 #50, 54,
11.10 #60, 62, 66,
choose 2 from 11.11

∞ Video Link: Chapter 11 Practice (15:59).
11.9 #50, 54,
11.10 #60, 62, 66, from 9e

★ TEACHER TIP
Students will begin selecting some of their own exercises to practice this week (if they haven't already done so). Check and see how they are doing!

WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 93**

▯ Materials: Introductory Chemistry

→ **REVIEW AND PRACTICE**
Continue exercises.

• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Don't forget to review your lesson notes!

# Science: Chemistry

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## Term 2

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 94**

▯ Materials: Introductory Chemistry

Prep: Read Teacher Tip

**➔ READ, NARRATE, & DISCUSS**

Ch.12.1-12.3

∞ Video Link: Chemical Bonding (6:17)

● Looking at the objectives at the start of the assigned sections, tell what you know about each topic.

● Discuss the Critical Thinking question in Sec.12-2.

★ TEACHER TIP

Note that students should use the objectives from the start of the assigned sections to assess their learning. Check that they are doing so and encourage them to message Homework Help, if they need support.

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 95**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.12.4-12.5

∞ Video Link: Understanding Ionic Solids (3:24)

● Looking at the objectives at the start of the assigned sections, tell what you know about each topic.

● Discuss the Critical Thinking question in Sec.12-5.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 4

⬚ **60m Chemistry Lessons - Lesson 96**

▯ Materials: Notebook or computer to compose

**➔ REPORT WRITING**

Prepare a composition to report on your energy research. This could be an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to effectively communicate what you have learned.

WEEK 5

⬚ **45m Chemistry Lessons - Lesson 97**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**

Ch.12.6

∞ Video Link: Lewis Structures (4:26)

● Looking at the objectives at the start of the assigned section, tell what you know about the topic.

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

# Science: Chemistry

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## Term 2

WEEK 5

**45m Chemistry Lessons - Lesson 98**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.12.7

∞ Video Link: Lewis Structures Continued (5:42)

● Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 5

**45m Chemistry Lessons - Lesson 99**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.12.8-12.9: If helpful, use the model kit to assemble some of the structures and observe their 3-dimensional geometry.

∞ Video Link: Understanding Covalent Structures (4:42)

● Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking? Do they know how to go back and find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-solving? If you are unsure about their progress, reach out through Contact Us.

WEEK 5

**45m Chemistry Lessons - Lesson 100**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**

Ch.12.10: If helpful, use the model kit to assemble some of the structures and observe their 3-dimensional geometry. You can try them first with single bonds and then compare them to the geometry when you add the double bond.

∞ Video Link: VSEPR Bonus (2:02)

● Tell what you know about the objectives at the start of this section.

**→ REVIEW AND PRACTICE**

Ch.12 Review and Suggested Exercises:
12.1 #6,
choose 3 from 12.2,
choose 2 from 12.3,
choose 3 from 12.4,
12.5 #46,
choose 3 from 12.6-7,
choose 3 from 12.9-10

∞ Video Link: Chapter 12 Practice (18:48).
12.3 #22,

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

## Term 2

12.5 #46,
12.6-7 #60, 66,
12.9-10 #82, from 9e

WEEK 5

⬚ **45m Chemistry Lessons - Lesson 101**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Continue exercises.

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 5

⬚ **60m Chemistry Lessons - Lesson 102**

▯ Materials: Introductory Chemistry

**→ CATCH UP & REVIEW**
Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 103**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.13.1-13.2
∞ Video Link: Boyle's Law (5:16)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 104**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.13.3
∞ Video Link: Charles' Law (3:52)

● Tell what you know about the objectives at the start of this section.
● Discuss the Critical Thinking question in Sec.13-3.

# Science: Chemistry

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## Term 2

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 105**

▯ Materials: Introductory Chemistry

**→ NOTE**
There is a typo in example 13.7 in edition 8. Avogadro's Law should have "n" in the denominator rather than "T."

**→ READ, NARRATE, & DISCUSS**
Ch.13.4
∞ Video Link: Avogadro's Law (2:31)

● Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 106**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.13.5
∞ Video Link: Ideal Gas Law (3:05)

● Tell what you know about the objectives at the start of this section.

★ TEACHER TIP
Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 107**
*Determining the Peroxide Content of a Commercial Antiseptic*

▯ Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**
Read the Introduction to Lab 9 and complete the introductory narration in your notebook.
∞ Video Link: Peroxide Lab (2:41)
∞ Image Link: Most Lab 9 materials

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 6 ⬚ **60m Chemistry Lessons - Lesson 108**
*Determining the Peroxide Content of a Commercial Antiseptic*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete Lab 9.

★ TEACHER TIP
Ask students to explain their lab to you!

# Science: Chemistry

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## Term 2

WEEK 7

⬚ **45m Chemistry Lessons - Lesson 109**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.13.6-13.7
∞ Video Link: Dalton's Law of Partial Pressures (7:19)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 7

⬚ **45m Chemistry Lessons - Lesson 110**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.13.8-13.10
∞ Video Link: Kinetic Theory (7:05)

● Tell what you know about the objectives at the start of these sections.
● Choose at least one Critical Thinking question in Sec.13-8 or 13-10 to discuss.

★ TEACHER TIP
The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 7

⬚ **45m Chemistry Lessons - Lesson 111**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.13 Review. Then choose exercises to complete as in previous chapters. Remember to try a selection of exercises to help discern where your understanding is secure and where you may need some additional review and practice. If the exercises seem comfortable, then continue. If they seem confusing or too challenging, review the associated reading(s) and practice more. The answers to even-numbered exercises are provided at the back of the book so you can check your understanding.

∞ Video Link: Chapter 13 Practice (41:23).
13.1 #12,
13.2 #22,
13.3 #36,
13.4 #42,
13.5 #52, 58,
13.6 #70, 74,
13.10 #96, 100, from 9e

! STUDENT REMINDER
Don't forget to review your lesson notes!

# Science: Chemistry

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## Term 2

WEEK 7 ⬚ **45m Chemistry Lessons - Lesson 112**

▯ Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**
Continue exercises.

• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 7 ⬚ **45m Chemistry Lessons - Lesson 113**

▯ Materials: High School Chemistry Lab Book

**➔ REPORT WRITING**
Begin writing a formal lab report for the candle enthalpy lab. It's been a while since you completed that lab, so it may be a challenge if you didn't keep detailed lab records. If this is the case, think about what would make your lab record more useful to you now and note this in your report.

WEEK 7 ⬚ **60m Chemistry Lessons - Lesson 114**

▯ Materials: High School Chemistry Lab Book

**➔ REPORT WRITING**
Complete your lab report. Review and edit it to make improvements. Use complete sentences, good grammar, third-person voice, and significant figures and scientific notation, as appropriate. As you read your report, imagine you are another scientist unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

WEEK 8 ⬚ **45m Chemistry Lessons - Lesson 115**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.14.1-14.2
∞ Video Link: Heats of Fusion and Vaporization (7:55)

• Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 8 ⬚ **45m Chemistry Lessons - Lesson 116**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ READ, NARRATE, & DISCUSS**

★ TEACHER TIP
Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

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## Term 2

Ch.14.3-14.4: If helpful, use the model kit to build water molecules. Then visualize water's intermolecular forces in solid, liquid, and gas states. What happens when salt ions are added to the liquid water model? Can you visualize it?
∞ Video Link: London Dispersion and Vapor Pressure (4:36)

- Tell what you know about the objectives at the start of these sections.
- Discuss the Critical Thinking question in Sec.14-3.

WEEK 8

⬚ **45m Chemistry Lessons - Lesson 117**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.14.5-14.6
∞ Video Link: Crystalline Solids (3:37)

- Try building the crystalline solids in Figure 14.14 with your model kit.
- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 8

⬚ **45m Chemistry Lessons - Lesson 118**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.14 Review. Then choose exercises to complete.

∞ Video Link: Chapter 14 Practice (18:41).
Act. #13, 17, 19,
14.1 #6 (also #8, Act. #20, 22),
14.2 #16,
14.3 #28, 30,
14.4 #38,
14.5 #46, from 9e

- Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 8

⬚ **45m Chemistry Lessons - Lesson 119**

*Separating a Mixture of Pigments*

Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**
Read the Introduction to Lab 10 and complete the introductory narration in your notebook.
∞ Video Link: Chromatography Lab (2:38)
∞ Image Link: Most Lab 10 materials

# Science: Chemistry

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## Term 2

WEEK 8

⬚ **60m Chemistry Lessons - Lesson 120**

*Separating a Mixture of Pigments*

Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

→ **LAB DAY**

Complete Lab 10.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 9

⬚ **45m Chemistry Lessons - Lesson 121**

Materials: Website link

→ **READ & PREPARE**

Explore the linked citizen science project you will participate in next term and prepare by discussing it with your teacher. You will use the nitrate and pH strips in your lab kit, as well as observe any stream invertebrates you find. Talk to your teacher about where you will collect your data for the project and make any necessary plans. Be sure to have a way to collect your water samples and possibly a dip net.

∞ Website Link: Clean Water Hub

∞ Website Link: Macroinvertebrate Gallery

∞ Website Link: Macroinvertebrate ID Guide

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

WEEK 9

⬚ **45m Chemistry Lessons - Lesson 122**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

→ **READ, NARRATE, & DISCUSS**

Ch.15.1-15.2: If helpful, use the model kit to build water molecules and demonstrate what happens when salt ions are added to the liquid water model. Build any of the molecules in Figures 15.3-5 and see how they interact with a water molecule.

∞ Video Link: Solutions (3:45)

● Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP

Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 9

⬚ **45m Chemistry Lessons - Lesson 123**

Materials: Introductory Chemistry

→ **READ, NARRATE, & DISCUSS**

Ch.15.3-15.4

∞ Video Link: Calculating Concentration (4:54)

! STUDENT REMINDER

Don't forget to review your lesson notes!

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## Term 2

● Tell what you know about the objectives at the start of these sections.

WEEK 9

⬚ **45m Chemistry Lessons - Lesson 124**

▯ Materials: Introductory Chemistry

➔ **READ, NARRATE, & DISCUSS**
Ch.15.5
∞ Video Link: Making Dilutions (3:41)

● Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 9

⬚ **45m Chemistry Lessons - Lesson 125**
*Observing Colligative Properties*

▯ Materials: High School Chemistry Lab Book and materials listed within

➔ **READ & NARRATE**
Read the Introduction to Lab 11 and complete the introductory narration in your notebook.
∞ Video Link: Colligative Properties Lab (4:51)
∞ Image Link: Most Lab 11 materials

WEEK 9

⬚ **60m Chemistry Lessons - Lesson 126**
*Observing Colligative Properties*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

➔ **LAB DAY**
Complete Lab 11.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 10

⬚ **45m Chemistry Lessons - Lesson 127**

▯ Materials: Introductory Chemistry

➔ **READ, NARRATE, & DISCUSS**
Ch.15.6
∞ Video Link: Solution Stoichiometry (3:59)

● Tell what you know about the objectives at the start of this section.
● Discuss the Critical Thinking question in Sec.15-6.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 2

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 128**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.15.7-15.8: If helpful, use the model kit to build the HCl and NaOH in section 15.7. Follow them during the reaction described to visualize what happens.
∞ Video Link: Stoichiometry and Normality (5:44)

● Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP
The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 129**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Ch.15 Review. Then choose exercises to begin.

∞ Video Link: Chapter 15 Practice (22:42).
15.3 #16, 24,
15.4 #40, 48,
15.5 #58, 61,
15.6 #66,
15.7 #72,
15.8 #88, from 9e

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 130**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**
Continue exercises.

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 131**
*Predicting a Neutralization Reaction*

Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**
Read the Introduction to Lab 12 and complete the introductory narration in your notebook.
∞ Video Link: Neutralization Lab (1:17)
∞ Image Link: Most Lab 12 materials

# Science: Chemistry

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## Term 2

WEEK 10 ⬚ **60m Chemistry Lessons - Lesson 132**

*Predicting a Neutralization Reaction*

Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete the Procedure for Lab 12.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 133**

Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned in this course. Review and edit the lab report before turning it in. While editing, notice your tense. Most of your report should be written in the past tense because it describes and explains something that already happened. The Introduction is often in the present tense, however. Sometimes, parts of the Discussion will be in the present or future tense. Consider what you are communicating to the reader with your tense.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 134**

Materials: High School Chemistry Lab Book

Introductory Chemistry

PREP: Read Teacher Tip

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly, whether they can explain their thinking and demonstrate a habit of problem-solving. If no relational growth is observed, consider why: do they need more practice, more support, less business, etc? Reach out through Contact Us, if you need help.

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 135**

Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 2

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 136**

▯ Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 137**

▯ Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 11 ⬚ **60m Chemistry Lessons - Lesson 138**

▯ Materials: High School Chemistry Lab Book

Introductory Chemistry

**→ REPORT WRITING & REVIEW**
Spend this week completing a lab report for Lab 11, then review what you have learned so far in this course.

WEEK 12 ⬚ **45m Chemistry Lessons - Lesson 139**

*Exam Week*

**→ EXAMS**
Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed.

WEEK 12 ⬚ **45m Chemistry Lessons - Lesson 140**

*Exam Week*

**→ EXAMS**
Answer question(s) related to course.

WEEK 12 ⬚ **45m Chemistry Lessons - Lesson 141**

*Exam Week*

**→ EXAMS**
Answer question(s) related to course.

  

# Science: Chemistry

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## Term 2

| | | |
|---|---|---|
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 142**<br>*Exam Week*<br><br>**➜ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 143**<br>*Exam Week*<br><br>**➜ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **60m Chemistry Lessons - Lesson 144**<br>*Exam Week*<br><br>**➜ EXAMS**<br>Answer question(s) related to course. | |

# Science: Chemistry

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## Term 3

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 145**

📕 Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➜ REMINDER**
Use the video lessons in whatever way is most helpful to you. You can turn on or off closed captions, speed them up or slow them down, use the pause button, etc. Some students prefer to watch the video lesson first and refer to the book as needed, some prefer to read the lesson first and watch the video as a summary or visual recap, and some don't like video lessons and don't use them at all. If you would like additional resources, you can find alternative videos in your Quick Links, as well as the Homework Help link to message Mrs. Merritt.

**➜ READ, NARRATE, & DISCUSS**
Ch.16.1-16.2
∞ Video Link: Acids and Bases (5:54)

• Tell what you know about the objectives at the start of these sections.
• Build the acids at the end of section 16.2 and identify the acidic protons.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

★ TEACHER TIP
The linked videos and the objectives at the start of each section are a great way for teachers to keep up with what students are learning, even when they can't read the book alongside students! If you prefer a slower, more conceptual approach without the problem-solving and lab tie-in, there are some alternatives in the Quick Links.

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WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 146**

📕 Materials: Introductory Chemistry

**➜ NOTE**
Logarithms are introduced in Sec.16.4. If you have not learned logarithms yet in math, that is okay. Only a conceptual understanding is needed at this time, which this text will teach you. As you read 16.4, notice that logarithms are the inverse operation of exponents (just as subtraction is the inverse operation of addition and division is the inverse operation of multiplication). With this idea of inverse operations, try to understand how the pH scale is related to the concentration of hydrogen cations.

**➜ READ, NARRATE, & DISCUSS**
Ch.16.3-16.4
∞ Video Link: The pH Scale (7:07)

• Tell what you know about the objectives at the start of these sections.
• Discuss the Critical Thinking question in Sec.16-4.

# Science: Chemistry

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## Term 3

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 147**

Materials: Introductory Chemistry

**→ NOTE**

Learners not yet ready to manipulate logarithms may skip Sec.16.5 and associated practice problems. If you want to go over the basics of logarithms as they relate to pH, consider watching the Math Break video:

∞ Video Link: Math Break: Logarithms (6:54)

**→ READ, NARRATE, & DISCUSS**

Ch.16.5-16.6: Visualize buffering by building at least 3 acetic acid molecules with your kit (the structure is at the end of section 16.2) and imagine putting these into water with the lesson. Since it is a weak acid, separate only 1 acidic proton. What happens when you add HCl, a strong acid, to this mixture? What happens when you add NaOH, a strong base?

∞ Video Link: Buffered Solutions (3:25)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 148**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Ch.16 Review. Then choose exercises to begin, skipping #41-54 and 57-58 if not ready to manipulate logarithms. You will have more time next week to complete these.

∞ Video Link: Chapter 16 Practice (34:42).

Act. #3,

16.1 #10, 14, 16,

16.2 #20, 25,

16.3 #32, 36,

16.4 #42, 44, 52, from 9e

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 1 ⬚ **45m Chemistry Lessons - Lesson 149**

*Analyzing the Content of Lemon Juice*

Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**

Read the Introduction to Lab 13 and complete the introductory narration in your notebook.

∞ Video Link: Lemon Juice Lab (2:59)

∞ Image Link: Most Lab 13 materials

# Science: Chemistry

Click THIS text or scan the QR code for links.

## Term 3

WEEK 1

☐ **60m Chemistry Lessons - Lesson 150**

Materials: Website link, container to collect water sample, nitrate and pH test strips, optional: dip net

PREP: Read Teacher Tip

**→ FIELD WORK**
Complete the first collection and analysis for your citizen science project: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.
∞ Website Link: Clean Water Hub
∞ Website Link: Macroinvertebrate Gallery
∞ Website Link: Macroinvertebrate ID Guide

★ TEACHER TIP
Ask students to tell about their citizen science project. What data did they collect? What is the purpose?

WEEK 2

☐ **45m Chemistry Lessons - Lesson 151**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ REVIEW AND PRACTICE**
Continue exercises.

• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

**• PLAN WEEKLY**
☐ nature walk - record observations
☐ science free read

★ TEACHER TIP
Use the objectives at the start of assigned sections to engage in discussion and note progress, even when students are working independently.

WEEK 2

☐ **45m Chemistry Lessons - Lesson 152**

Materials: Current Events link

Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below:
∞ Website Link: Science News for Students
∞ Website Link: Chemistry at Science Daily
∞ Website Link: Chemical & Engineering News

**→ READ, NARRATE, & DISCUSS**
Ch.17.1-17.2
∞ Video Link: Collision Theory (3:33)

• Tell what you know about the objectives at the start of these sections.
• Choose at least one Critical Thinking question in Sec.17-1 or 17-2.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

# Science: Chemistry

[Click THIS text or scan the QR code for links.]

## Term 3

WEEK 2

⬚ **45m Chemistry Lessons - Lesson 153**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.17.3-17.4
∞ Video Link: Chemical Equilibrium (2:52)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Don't forget to review your lesson notes!

WEEK 2

⬚ **45m Chemistry Lessons - Lesson 154**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.17.5-17.6
∞ Video Link: The Equilibrium Constant (5:05)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 2

⬚ **45m Chemistry Lessons - Lesson 155**
*Analyzing the Content of Lemon Juice*

▯ Materials: High School Chemistry Lab Book and materials listed within

**➔ LAB DAY**
Begin Lab 13.

WEEK 2

⬚ **60m Chemistry Lessons - Lesson 156**
*Analyzing the Content of Lemon Juice*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**➔ LAB DAY**
Complete Lab 13.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 157**

▯ Materials: High School Chemistry Lab Book

**➔ REPORT WRITING**
Write a lab report for Lab 13. Review and edit before turning it in. Use complete sentences, good grammar, third-person voice, correct tense, significant figures, and scientific notation, as appropriate. Remember to read the report as if you were another scientist

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 158**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.17.7

∞ Video Link: Le Châtelier's Principle (4:31)

● Tell what you know about the objectives at the start of this section.

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 159**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**

Ch.17.8-17.9

∞ Video Link: Using K (4:07)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

Don't forget to review your lesson notes!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 160**

▯ Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Ch.17 Review. Then choose exercises to begin. You will have more time next week to complete these.

∞ Video Link: Chapter 17 Practice (25:05).

17.5 #18, 22,

17.6 #26,

17.7 #35, 40,

17.8 #46, 50,

17.9 #60, 64, 68, from 9e

! STUDENT REMINDER

Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 3

⬚ **45m Chemistry Lessons - Lesson 161**

*Analyzing Reaction Kinetics*

▯ Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**

Read the Introduction to Lab 14 and complete the introductory

## Term 3

narration in your notebook.
∞ Video Link: Kinetics (1:05)
∞ Image Link: Most Lab 14 materials

WEEK 3

⬚ **60m Chemistry Lessons - Lesson 162**

*Analyzing Reaction Kinetics*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**➔ LAB DAY**
Complete Lab 14.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 163**

▯ Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**
Continue exercises.

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 164**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ READ, NARRATE, & DISCUSS**
Ch.18.1-18.2
∞ Video Link: Reduction and Oxidation (5:03)

● Tell what you know about the objectives at the start of these sections.
● Discuss the Critical Thinking question in Sec.18-2.

★ TEACHER TIP
Don't forget to check that students are using the objectives at the start of assigned sections to assess their learning. Encourage them to message Homework Help, if they need support.

WEEK 4

⬚ **45m Chemistry Lessons - Lesson 165**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.18.3
∞ Video Link: Redox between Nonmetals (3:57)

● Tell what you know about the objectives at the start of this section.

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WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 166**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.18.4
∞ Video Link: Balancing Redox Reactions (4:46)

● Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 4 ⬚ **45m Chemistry Lessons - Lesson 167**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.18.5-18.7
∞ Video Link: Electrochemistry (7:36)

● Discuss the Critical Thinking question in Sec.18-7.

● Tell what you know about the objectives at the start of these sections.

WEEK 4 ⬚ **60m Chemistry Lessons - Lesson 168**

Materials: Introductory Chemistry

**→ CATCH UP & REVIEW**
Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 5 ⬚ **45m Chemistry Lessons - Lesson 169**

Materials: Introductory Chemistry

**→ NOTE**
This section is more about electrolysis as the reverse reaction to that in the galvanic cell, which was already touched on in the last reading and in our video. Read it if interested, but you do not need to spend a lot of time with it.

**→ READ, NARRATE, & DISCUSS**
Ch.18.8.

**→ REVIEW AND PRACTICE**
Ch.18 Review. Then choose exercises to begin.

∞ Video Link: Chapter 18 Practice (32:42).

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

## Term 3

18.1 #6,
18.2 #14, 18, 22,
18.3 #34, 36,
18.4 #42, 46,
18.5 #54, from 9e

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WEEK 5

⬚ **45m Chemistry Lessons - Lesson 170**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ REVIEW AND PRACTICE**
Continue exercises.

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

★ TEACHER TIP
Learners encounter a lot more technical content at this disciplinary stage. If they do not remember or fully understand every detail, that is okay. Can they talk about it and explain their thinking? Do they know how to go back and find what they need if they need it? Are they able to lean into the uncertainty with curiosity and a habit of problem-solving? If you are unsure about their progress, reach out through Contact Us.

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WEEK 5

⬚ **45m Chemistry Lessons - Lesson 171**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.19.1-19.2
∞ Video Link: Radioactivity (6:32)

● Tell what you know about the objectives at the start of these sections.
● Discuss the Critical Thinking question in Sec.19-1.

! STUDENT REMINDER
Don't forget to review your lesson notes!

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WEEK 5

⬚ **45m Chemistry Lessons - Lesson 172**

▯ Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.19.3-19.4
∞ Video Link: Half-Life (3:16)

● Tell what you know about the objectives at the start of these sections.

∞ Video Link: How does radiocarbon dating work? (2:10)

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

# Science: Chemistry

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## Term 3

WEEK 5

⬚ **45m Chemistry Lessons - Lesson 173**

*Producing Hydrogen and Oxygen from Water*

> ▯ Materials: High School Chemistry Lab Book and materials listed within

**→ READ & NARRATE**

Read the Introduction to Lab 15 and complete the introductory narration in your notebook.

∞ Video Link: Electrolysis Lab (3:36)

∞ Image Link: Most Lab 15 materials

WEEK 5

⬚ **60m Chemistry Lessons - Lesson 174**

*Producing Hydrogen and Oxygen from Water*

> ▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**

Complete Lab 15.

★ TEACHER TIP

Ask students to explain their lab to you!

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 175**

> ▯ Materials: Website link

Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**

What interests you? Find out what chemistry can teach you about that interest by searching for it or browse to discover something new at one of the current event resources below:

∞ Website Link: Science News for Students

∞ Website Link: Chemistry at Science Daily

∞ Website Link: Chemical & Engineering News

**→ READ, NARRATE, & DISCUSS**

Ch.19.5

! STUDENT REMINDER

Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**

⬚ nature walk - record observations

⬚ science free read

WEEK 6

⬚ **45m Chemistry Lessons - Lesson 176**

> ▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**

Ch.19.6-19.10

∞ Video Link: Nuclear Energy (4:54)

★ TEACHER TIP

The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

Click THIS text or scan the QR code for links.

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● Tell what you know about the objectives at the start of these sections.
● Discuss the Critical Thinking question in Sec.19-8.

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 177**

Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**
Ch.19 Review. Then choose exercises to begin.

∞ Video Link: Chapter 19 Practice (11:14).
Act. #2,
19.1 #22, 28,
19.3 #40, 42, from 9e

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 178**

Materials: Introductory Chemistry

**➔ REVIEW AND PRACTICE**
Continue exercises.

● Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing?

WEEK 6 ⬚ **45m Chemistry Lessons - Lesson 179**

Materials: Introductory Chemistry

**➔ CATCH UP & REVIEW**
Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 6 ⬚ **60m Chemistry Lessons - Lesson 180**

Materials: Website link, container to collect water sample, nitrate and pH test strips, optional: dip net

PREP: Read Teacher Tip

**➔ FIELD WORK**
Complete the second collection and analysis for your citizen science project: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.

★ TEACHER TIP
Ask students to tell about their citizen science project. Was their data different today or the same as the first collection? Did they notice anything different at the site?

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Click THIS text or scan the QR code for links.

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∞ Website Link: Clean Water Hub
∞ Website Link: Macroinvertebrate Gallery
∞ Website Link: Macroinvertebrate ID Guide

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WEEK 7

⬚ **45m Chemistry Lessons - Lesson 181**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.20.1-20.2: Use your model kit!
∞ Video Link: Alkanes (4:51)

● Tell what you know about the objectives at the start of these sections. Use your model kit as much as is helpful throughout this chapter!

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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WEEK 7

⬚ **45m Chemistry Lessons - Lesson 182**

Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.20.3-20.4: Use your model kit!
∞ Video Link: Branched Alkanes (9:55)

● Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP
The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

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WEEK 7

⬚ **45m Chemistry Lessons - Lesson 183**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.20.5-20.7
∞ Video Link: Alkenes and Alkynes (7:29)

● Tell what you know about the objectives at the start of these sections.
● Discuss the Critical Thinking question in Sec.20-5.

! STUDENT REMINDER
Don't forget to review your lesson notes!

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WEEK 7

⬚ **45m Chemistry Lessons - Lesson 184**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.20.8-20.9
∞ Video Link: Rings (8:22)

● Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

# Science: Chemistry

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## Term 3

WEEK 7

⬚ **45m Chemistry Lessons - Lesson 185**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.20.10-20.12
∞ Video Link: Functional Groups Part 1 (4:52)

• Tell what you know about the objectives at the start of these sections.

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WEEK 7

⬚ **60m Chemistry Lessons - Lesson 186**

*Hydrolyzing Esters*

▯ Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ READ & NARRATE**
Read the Introduction to Lab 16 and complete the introductory narration in your notebook.
∞ Video Link: Soap Lab (5:30)
∞ Image Link: Most Lab 16 materials

★ TEACHER TIP
Ask students to explain their lab to you!

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WEEK 8

⬚ **45m Chemistry Lessons - Lesson 187**

▯ Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.20.13-20.14
∞ Video Link: Functional Groups Part 2 (4:06)

• Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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WEEK 8

⬚ **45m Chemistry Lessons - Lesson 188**

▯ Materials: Introductory Chemistry

PREP: Read Teacher Tip

**→ READ, NARRATE, & DISCUSS**
Ch.20.15-20.16: Use your model kit!
∞ Video Link: Functional Groups Part 3 (8:00)

• Tell what you know about the objectives at the start of these sections.

★ TEACHER TIP
The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

# Science: Chemistry

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## Term 3

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|---|---|---|
| WEEK 8 | ⬚ **45m Chemistry Lessons - Lesson 189**<br><br>▯ Materials: Introductory Chemistry<br><br>**→ NOTE**<br>Review and practice for the last two chapters of the book are largely about applying the rules and definitions from the chapter. Look back at the rules and definitions in the text as much as you need to practice. Use Homework Help if you need help.<br><br>**→ REVIEW AND PRACTICE**<br>Ch.20 Review. Then choose exercises to begin. | ! STUDENT REMINDER<br>Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset! |
| WEEK 8 | ⬚ **45m Chemistry Lessons - Lesson 190**<br><br>▯ Materials: Introductory Chemistry<br><br>**→ REVIEW AND PRACTICE**<br>Continue exercises.<br><br>• Remember to self-assess before moving to the next chapter. Do the exercises make sense, or do they feel confusing? | ! STUDENT REMINDER<br>Don't forget to review your lesson notes! |
| WEEK 8 | ⬚ **45m Chemistry Lessons - Lesson 191**<br>*Hydrolyzing Esters*<br><br>▯ Materials: High School Chemistry Lab Book and materials listed within<br><br>**→ LAB DAY**<br>Begin Lab 16. Note that the mixing phase of this experiment can take 1-2 hours, depending on temperature and mixing efficiency. | |
| WEEK 8 | ⬚ **60m Chemistry Lessons - Lesson 192**<br>*Hydrolyzing Esters*<br><br>▯ Materials: High School Chemistry Lab Book and materials listed within<br><br>PREP: Read Teacher Tip<br><br>**→ LAB DAY**<br>Continue Lab 16. | ★ TEACHER TIP<br>Ask students to explain their lab to you! |
| WEEK 9 | ⬚ **45m Chemistry Lessons - Lesson 193**<br>*Hydrolyzing Esters*<br><br>▯ Materials: High School Chemistry Lab Book and materials listed within<br><br>**→ LAB DAY** | ! STUDENT REMINDER<br>Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.<br><br>**• PLAN WEEKLY**<br>⬚ nature walk - record observations<br>⬚ science free read |

## Term 3

Complete Lab 16.

WEEK 9 **45m Chemistry Lessons - Lesson 194**

Materials: Introductory Chemistry

**→ REVIEW AND PRACTICE**

Continue exercises.

WEEK 9 **45m Chemistry Lessons - Lesson 195**

Materials: High School Chemistry Lab Book

**→ REPORT WRITING**

Write a lab report for Lab 16. Review and edit before turning it in. Use complete sentences, good grammar, third-person voice, correct tense, and significant figures and scientific notation, as appropriate. Remember to read as if you were another scientist who is unfamiliar with the experiment. Does the report make sense? Would you be able to repeat the experiment using the lab report? If not, edit the report to make it so.

WEEK 9 **45m Chemistry Lessons - Lesson 196**

Materials: Introductory Chemistry

**→ CATCH UP & REVIEW**

Complete any unfinished work, then do a cumulative review of all you have learned. As you look back over your notes and narrate objectives from the chapters, reflect on whether the ideas make sense or seem confusing. If anything is confusing, read back over the associated section(s), try some exercises, and/or use Homework Help.

WEEK 9 **45m Chemistry Lessons - Lesson 197**

Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**

Ch.21.1-21.4: Build the amino acid shown in section 21.2 (you can use any single atom that you want to represent the R group). Identify the functional groups on this amino acid. Then you can play with building some of the R groups shown in Figure 21.2. Very large structures are needed to demonstrate secondary and tertiary structure. How could you visualize these?

∞ Video Link: Protein Structure (6:09)

- Tell what you know about the objectives at the start of these sections.

! STUDENT REMINDER

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# Science: Chemistry

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## Term 3

**WEEK 9** ⬚ **60m Chemistry Lessons - Lesson 198**

*Polymerizing with Borax*

> Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**➔ LAB DAY**
Complete Lab 17.
∞ Video Link: Silly Putty Lab (4:37)
∞ Image Link: Most Lab 17 materials

★ TEACHER TIP
Ask students to explain their lab to you!

**WEEK 10** ⬚ **45m Chemistry Lessons - Lesson 199**

> Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.21.5-21.6
∞ Video Link: Protein Function (3:46)

- Tell what you know about the objectives at the start of these sections.
- The enzyme-substrate mechanism described in your book is very rudimentary. To learn about the more commonly accepted, dynamic model, watch the linked Induced Fit video.

∞ Video Link: Induced Fit (6:40)

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

**WEEK 10** ⬚ **45m Chemistry Lessons - Lesson 200**

> Materials: Introductory Chemistry

PREP: Read Teacher Tip

**➔ READ, NARRATE, & DISCUSS**
Ch.21.7: Use your model kit to build a carbohydrate. Compare and contrast with the amino acid.
∞ Video Link: Carbohydrates (5:01)

- Tell what you know about the objectives at the start of this section.

★ TEACHER TIP
The videos and section objectives are a great way to keep up and engage with students, even when they are working independently. Your positive support matters!

**WEEK 10** ⬚ **45m Chemistry Lessons - Lesson 201**

> Materials: Introductory Chemistry

**➔ READ, NARRATE, & DISCUSS**
Ch.21.8
∞ Video Link: Nucleic Acids (4:41)

- Tell what you know about the objectives at the start of this section.

! STUDENT REMINDER
Don't forget to review your lesson notes!

## Term 3

● Can you use your model kit to build a nucleic acid? What features make something a nucleic acid? How else could you model this large molecule?

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 202**

*Chelating with Sodium Alginate*

> Materials: High School Chemistry Lab Book and materials listed within

**→ LAB DAY**
Begin Lab 18.
∞ Video Link: Dessert Caviar Lab (5:30)
∞ Image Link: Most Lab 18 materials

WEEK 10 ⬚ **45m Chemistry Lessons - Lesson 203**

> Materials: Introductory Chemistry

**→ READ, NARRATE, & DISCUSS**
Ch.21.9
∞ Video Link: Lipids (6:15)

● Tell what you know about the objectives at the start of this section.
● Tell how structure produces the common property shared by all lipids.

**→ REVIEW AND PRACTICE**
Review and practice for the last two chapters of the book are largely about applying the rules and definitions from the chapter. Look back at the rules and definitions in the text as much as you need to practice.
Ch.21 Review. Then choose exercises to begin.

! STUDENT REMINDER
Doing science is like solving a puzzle. Adjust your pace and use the tools provided to maintain that mindset!

WEEK 10 ⬚ **60m Chemistry Lessons - Lesson 204**

*Chelating with Sodium Alginate*

> Materials: High School Chemistry Lab Book and materials listed within

PREP: Read Teacher Tip

**→ LAB DAY**
Complete Lab 18.

★ TEACHER TIP
Ask students to explain their lab to you!

WEEK 11 ⬚ **45m Chemistry Lessons - Lesson 205**

> Materials: Introductory Chemistry

Website link, container to collect water sample, nitrate and pH test strips, optional: dip net

**→ WRAP UP & REVIEW**

! STUDENT REMINDER
Don't forget to use Homework Help in your Quick Links if you need any help or to offer a thought for discussion.

● **PLAN WEEKLY**

# Science: Chemistry

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## Term 3

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| | Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. Remember that your data collection is as follows: collect a water sample from your chosen location, dip each test strip into the sample in turn, compare the test strip to the color key on the package to record your data, then make any observations regarding the local geography, plant and animal life, etc.<br>As part of your citizen science project, create a poster or graphic summarizing the water health at your test site. Remember that you want to think about how to communicate your results effectively! You can do this on paper or in a digital program, like Canva, turning it in as part of your exam.<br>∞ Website Link: Clean Water Hub<br>∞ Website Link: Macroinvertebrate Gallery<br>∞ Website Link: Macroinvertebrate ID Guide<br>∞ Video Link: Meringue Lab (4:59)<br>∞ Image Link: Most Lab 19 materials | ⬚ nature walk - record observations<br>⬚ science free read |
| WEEK 11 | ⬚ **45m Chemistry Lessons - Lesson 206**<br>Materials: Introductory Chemistry<br>Citizen Science Project<br><br>PREP: Read Teacher Tip<br><br>**→ WRAP UP & REVIEW**<br>Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. | ★ TEACHER TIP<br>Ask students to explain their lab to you! |
| WEEK 11 | ⬚ **45m Chemistry Lessons - Lesson 207**<br>Materials: Introductory Chemistry<br>Citizen Science Project<br><br>PREP: Read Teacher Tip<br><br>**→ WRAP UP & REVIEW**<br>Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. | ★ TEACHER TIP<br>Ask students to tell about their citizen science project. Was the data different today or the same? What does that mean? Why do they think that? Did they notice anything different at the site this week? |
| WEEK 11 | ⬚ **45m Chemistry Lessons - Lesson 208**<br>Materials: Introductory Chemistry<br>Citizen Science Project<br><br>PREP: Read Teacher Tip<br><br>**→ WRAP UP & REVIEW** | ★ TEACHER TIP<br>When student(s) take exams next week, evaluate what they recall, but more importantly, whether they can explain their thinking and demonstrate a habit of problem-solving.  If no relational growth is observed, consider why: do they need more practice, more support, less |

# Science: Chemistry

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## Term 3

| | | |
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| | Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. | business, etc? Reach out through Contact Us, if you need help. |
| WEEK 11 | ⬚ **45m Chemistry Lessons - Lesson 209**<br>Materials: Introductory Chemistry<br>Citizen Science Project<br><br>**➔ WRAP UP & REVIEW**<br>Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. | |
| WEEK 11 | ⬚ **60m Chemistry Lessons - Lesson 210**<br>Materials: Introductory Chemistry<br>Citizen Science Project<br><br>**➔ WRAP UP & REVIEW**<br>Spend this week completing Lab 19, Ch.21 exercises, your third Citizen Science collection (and data upload), and reviewing what you have learned in this course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 211**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course. Complete about three questions per day or as directed by your teacher. You may use the reference material inside the covers of your book as needed. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 212**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 213**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 214**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to course. | |

# Science: Chemistry

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## Term 3

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| WEEK 12 | ⬚ **45m Chemistry Lessons - Lesson 215**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |
| WEEK 12 | ⬚ **60m Chemistry Lessons - Lesson 216**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to course. | |

# Science: Chemistry

## Examination

### Term 1

**Chemistry Lessons**

· Day 1: Choose 2 of the following from Introductory Chemistry:
● #8, #12, #24, #26 from Ch.4-5 Cumulative Rev

· Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous day's questions as appropriate based on material covered):
● #12, #16, #10, #18 from Ch.6-7 Cumulative Rev

· Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous days' questions as appropriate based on material covered):
● #2, #14, #18, #20 from Ch.8-9 Cumulative Rev

· Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.

· Day 5: Explain an example of chemistry in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

### Term 2

**Chemistry Lessons**

· Day 1: Choose 2 of the following from Introductory Chemistry (substitute previous term's questions as appropriate based on material covered):
● #2, #14, #20, #24 from Ch.10-12 Cumulative Rev

· Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
● #28, #34, #48 from Ch.10-12 Cumulative Rev, #4 from Ch.13-15 Cumulative Rev

· Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):
● #14, #22, #32, #36 from Ch.13-15 Cumulative Rev

· Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.

· Day 5: Explain an example of chemistry in the world around you that you noticed this term. This might be from your project work, current events, nature walks, or your free read. Describe any questions this made you think of.

## Term 3

### Chemistry Lessons

· Day 1: Choose 2 of the following from Introductory Chemistry (substitute previous term's questions as appropriate based on material covered):

● #2, #10, #14, #22 from Ch.16-17 Cumulative Rev

· Day 2: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):

● #84, #94 from Ch.18, #92, #96 from Ch.19

· Day 3: Choose 2 of the following from Introductory Chemistry (substitute previous questions as appropriate based on material covered):

● #118, #134 from Ch.20, #14, #42 from Ch.21

· Day 4: Explain one of your laboratory exercises from this term and what you learned from it. Describe any questions this made you think of.

· Day 5: Submit your citizen science poster or graphic, supplemented with any stories or explanations that are helpful. Convey any questions this project made you think of.